Лабораторная работа №13

Настройка NFS

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Группа: НФИбд 02–23

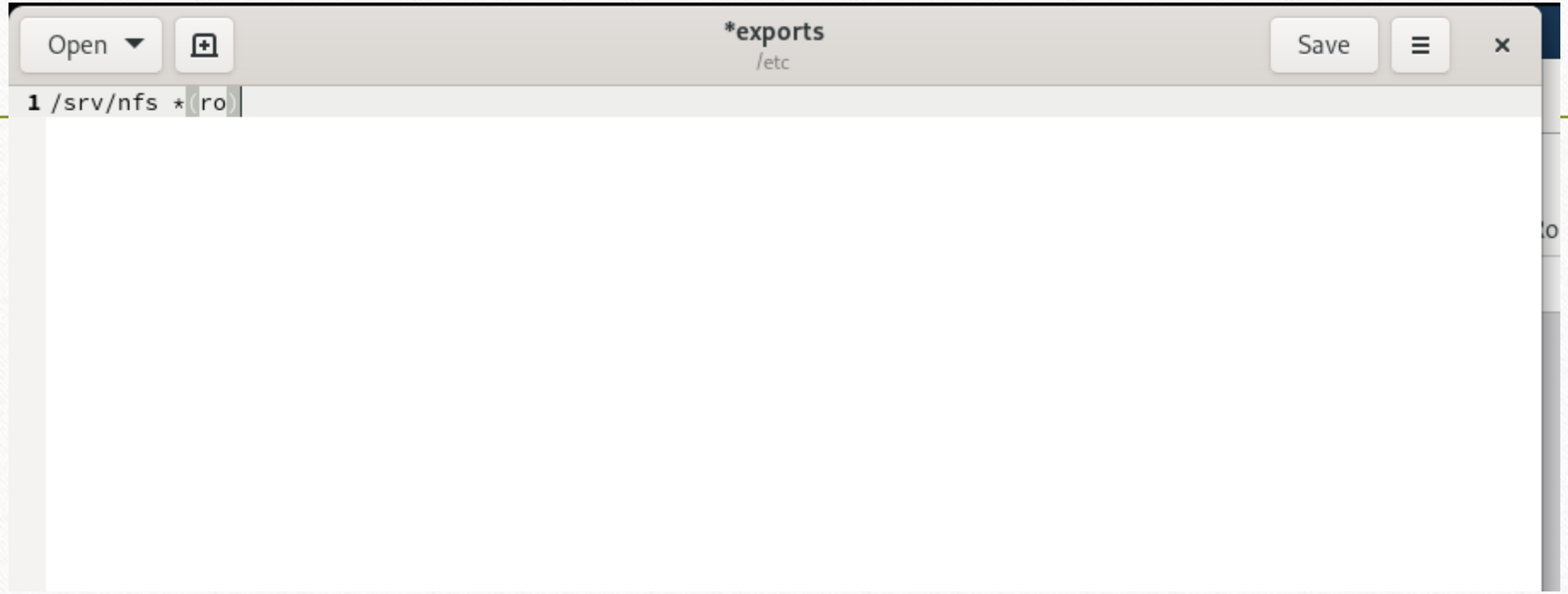
дисциплина: **Администрирование сетевых подсистем (Lab 13)**

Цель работы

- Целью данной работы является приобретение навыков настройки сервера NFS для удалённого доступа к ресурсам.

Рис. 1.1. Установка на сервере программного обеспечения nfs-utils.

```
[vagrant@server ~]$ dnf -y install nfs-utils
Error: This command has to be run with superuser privileges (under the root user on most systems).
[vagrant@server ~]$ sudo -i
[root@server ~]# dnf -y install nfs-utils
Last metadata expiration check: 1:21:33 ago on Thu 12 Mar 2026 09:05:56 AM UTC.
Dependencies resolved.
================================================================================
 Package                Arch        Version                   Repository   Size
================================================================================
Installing:
 nfs-utils              x86_64      1:2.5.4-38.el9_7.3        baseos      434 k
Upgrading:
 libipa_hbac            x86_64      2.9.7-4.el9_7.1           baseos       33 k
 libldb                 x86_64      4.22.4-12.el9_7           baseos      180 k
 libsmbclient           x86_64      4.22.4-12.el9_7           baseos       73 k
 libsss_certmap         x86_64      2.9.7-4.el9_7.1           baseos       88 k
 libsss_idmap           x86_64      2.9.7-4.el9_7.1           baseos       39 k
 libsss_nss_idmap       x86_64      2.9.7-4.el9_7.1           baseos       43 k
 libsss_sudo            x86_64      2.9.7-4.el9_7.1           baseos       33 k
 libtalloc              x86_64      2.4.3-1.el9               baseos       33 k
 libtdb                 x86_64      1.4.13-1.el9              baseos       53 k
 libtevent              x86_64      0.16.2-1.el9              baseos       50 k
 libwbclient            x86_64      4.22.4-12.el9_7           baseos       42 k
 samba-client-libs      x86_64      4.22.4-12.el9_7           baseos      5.3 M
 samba-common           noarch      4.22.4-12.el9_7           baseos      173 k
 samba-common-libs      x86_64      4.22.4-12.el9_7           baseos      104 k
 sssd                   x86_64      2.9.7-4.el9_7.1           baseos       25 k
 sssd-ad                x86_64      2.9.7-4.el9_7.1           baseos      217 k
 sssd-client            x86_64      2.9.7-4.el9_7.1           baseos      158 k
 sssd-common            x86_64      2.9.7-4.el9_7.1           baseos      1.6 M
 sssd-common-pac        x86_64      2.9.7-4.el9_7.1           baseos       94 k
 sssd-ipa               x86_64      2.9.7-4.el9_7.1           baseos      286 k
 sssd-kcm               x86_64      2.9.7-4.el9_7.1           baseos      107 k
 sssd-krb5              x86_64      2.9.7-4.el9_7.1           baseos       70 k
 sssd-krb5-common       x86_64      2.9.7-4.el9_7.1           baseos       92 k
 sssd-ldap              x86_64      2.9.7-4.el9_7.1           baseos      159 k
 sssd-proxy             x86_64      2.9.7-4.el9_7.1           baseos       70 k
```


Open
*exports
/etc
Save
1 /srv/nfs *(ro)

root@server:/etc

```
[root@server etc]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
      Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
      Output information may be incomplete.
systemd      1                          root   63u     IPv4              45663       0t0        UD
P *:sunrpc
systemd      1                          root   65u     IPv6              45677       0t0        UD
P *:sunrpc
avahi-dae  637                         avahi   12u     IPv4              19843       0t0        UD
P *:mdns
avahi-dae  637                         avahi   13u     IPv6              19844       0t0        UD
P *:mdns
avahi-dae  637                         avahi   14u     IPv4              19845       0t0        UD
P *:60877
avahi-dae  637                         avahi   15u     IPv6              19846       0t0        UD
P *:52703
chronyd    658                        chrony    5u     IPv4              19840       0t0        UD
P localhost:323
chronyd    658                        chrony    6u     IPv6              19841       0t0        UD
P localhost:323
chronyd    658                        chrony    7u     IPv4              19842       0t0        UD
P *:ntp
NetworkMa  928                          root   30u     IPv4              22111       0t0        UD
P server.user.net:bootpc->_gateway:bootps
NetworkMa  928  931 gmain               root   30u     IPv4              22111       0t0        UD
P server.user.net:bootpc->_gateway:bootps
NetworkMa  928  932 gdbus               root   30u     IPv4              22111       0t0        UD
P server.user.net:bootpc->_gateway:bootps
rpcbind   7310                           rpc    5u     IPv4              45663       0t0        UD
P *:sunrpc
rpcbind   7310                           rpc    7u     IPv6              45677       0t0        UD
P *:sunrpc
rpc.statd 7313                       rpcuser    7u     IPv4              54670       0t0        UD
P *:50942
rpc.statd 7313                       rpcuser    9u     IPv4              54664       0t0        UD
P localhost:708
```

```
[root@client ~]# mount server.user.net:/srv/nfs /mnt/nfs
mount: /mnt/nfs: bad option; for several filesystems (e.g. nfs, cifs) you might need a /sbin/mount
.<type> helper program.
[root@client ~]# mount
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,seclabel)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=4096k,nr_inodes=448710,mode=755,inode64)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,seclabel,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,seclabel,gid=5,mode=620,ptmxmode=000)
tmpfs on /run type tmpfs (rw,nosuid,nodev,seclabel,size=729744k,nr_inodes=819200,mode=755,inode64)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,seclabel,nsdelegate,memory
_recursiveprot)
pstore on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime,seclabel)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
/dev/sda1 on / type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
selinuxfs on /sys/fs/selinux type selinuxfs (rw,nosuid,noexec,relatime)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=29,pgrp=1,timeout=0,minproto=5,m
axproto=5,direct,pipe_ino=19574)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime,seclabel)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,relatime,seclabel,pagesize=2M)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime,seclabel)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime,seclabel)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
none on /run/credentials/systemd-sysctl.service type ramfs (ro,nosuid,nodev,noexec,relatime,seclab
el,mode=700)
```

Rocky

Right Ctrl

Open Save

*fstab
/etc

```
#
# /etc/fstab
# Created by anaconda on Fri Jul 26 13:25:41 2024
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=c84cce45-9089-48d9-9617-2f1d1bd45fdd /                       xfs     defaults        0 0
/swapfile none swap defaults 0 0
#VAGRANT-BEGIN
# The contents below are automatically generated by Vagrant. Do not modify.
#VAGRANT-END

/var/www /srv/nfs/www none bind 0 0
```

Open ▾ | fstab /etc | Save

```
#
# /etc/fstab
# Created by anaconda on Fri Jul 26 13:25:41 2024
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
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UUID=c84cce45-9089-48d9-9617-2f1d1bd45fdd /                       xfs     defaults        0 0
/swapfile none swap defaults 0 0
#VAGRANT-BEGIN
# The contents below are automatically generated by Vagrant. Do not modify.
#VAGRANT-END

/var/www /srv/nfs/www none bind 0 0
/home/user/common /srv/nfs/home/vagrant none bind 0 0
```

Saving file "/etc/fstab"... | Plain Text ▾ | Tab Width: 8 ▾ | Ln 19, Col 54 ▾ | INS

Open ▾ | nfs.sh /vagrant/provision/server | Save

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install nfs-utils
echo "Copy configuration files"
cp -R /vagrant/provision/server/nfs/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-service nfs --permanent
firewall-cmd --add-service mountd --add-service rpc-bind --permanent
firewall-cmd --reload
echo "Tuning SELinux"
mkdir -p /srv/nfs
semanage fcontext -a -t nfs_t "/srv/nfs(/.*)?"
restorecon -vR /srv/nfs
echo "Mounting dirs"
mkdir -p /srv/nfs/www
mount -o bind /var/www /srv/nfs/www
echo "/var/www /srv/nfs/www none bind 0 0" >> /etc/fstab
mkdir -p /srv/nfs/home/user
mkdir -p -m 700 /home/user/common
chown user:user /home/user/common
mount -o bind /home/user/common /srv/nfs/home/user
echo "/home/user/common /srv/nfs/home/user none bind 0 0" >> /etc/fstab
echo "Start nfs service"
systemctl enable nfs-server
systemctl start nfs-server
systemctl restart firewalld
```

sh ▾ Tab Width: 8 ▾ Ln 19, Col 57 ▾ INS

Вывод

- В ходе выполнения лабораторной работы были приобретены навыки настройки сервера NFS для удалённого доступа к ресурсам.

Спасибо за внимание!